

[DC] Immersive Analytics for Understanding Ecosystem Services Tradeoffs

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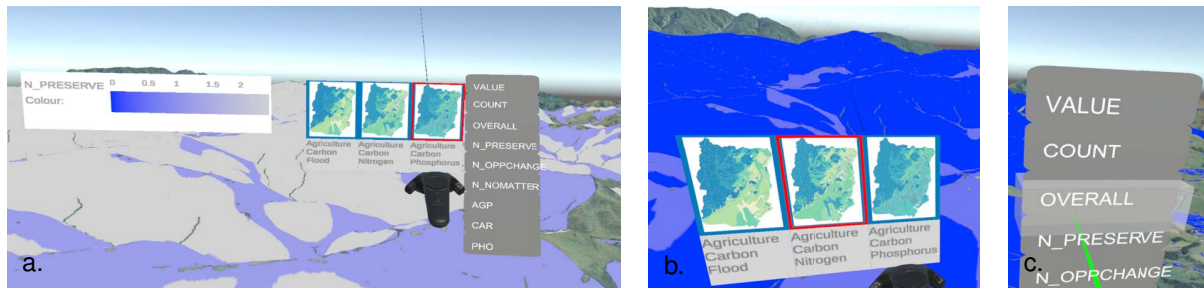


Figure 1: Interface from *Immersive ESS Visualizer* shows the layer chooser, attribute chooser and legend. The selected tradeoff layer among agriculture, carbon, and phosphorus is outlined in red Fig. 1a. The layer chooser selection is shown in Fig. 1b and attribute menu with overall selected in Fig. 1c. The tradeoff layer is overlaid on a map of the Mangatarere catchment area, NZ.

ABSTRACT

Existing immersive systems for analysing geospatial data relating to ecosystem services are not designed for all groups involved with land use decision making. Land management scientists have different requirements compared to non-experts as the tasks they perform are different. Land use decision making needs better tools for assisting the analysis and exploration of land use decisions, and their effect on ecosystem services. In this research, a user centred design process is applied for developing and evaluating an immersive VR visualization tool to assist with better decision making around land use. Interviews with experts found issues with how their current tool presents analysis results, and problems with communicating their results to stakeholders. A literature review found no pre-existing immersive VR systems specifically for analysing tradeoffs among ecosystem services.

Index Terms: Human-centered computing—Visualization—Visualization techniques; Human-centered computing—Visualization—Visualization design and evaluation methods Applied computing—Physical sciences and engineering—Earth and atmospheric sciences

1 INTRODUCTION

Ecosystem services are the benefits to humans provided by ecosystems through their internal natural processes. Land use is how humans manage, alter and conserve the ecosystem services and goods provided by the land [5]. Changing land use to improve one ecosystem service can affect other ecosystem services so a potential land use change needs to be analysed before its implementation. Analysing ecosystem services tradeoffs [5] helps prevent undesirable reductions in ecosystem services that need to be preserved while ensuring that the proposed land use changes are effective for their intended purpose.

The Land Utilisation Capability Indicator (LUCI) models ecosystem services including flood risk, flood mitigation, agricultural production, nitrogen, phosphorus, erosion, carbon sequestration, and biodiversity [3, 4]. LUCI takes input data from an elevation grid,

stream networks, annual rainfall, annual evaporation, spring additions and abstractions, land cover, and soil data. The analysis can be performed at a farm, catchment or landscape scale, and decisions can be made based on visualizing the output data. Data sources include soil maps and annual rainfall. These are represented as surfaces that overlay the mapped region being analysed.

Existing land use analysis tools are non-immersive and tradeoff analysis requires a large amount of spatial information to be analysed which can be difficult to compare and communicate to non-experts. A VR visualization tool could help land management scientists to better understand the land use categorized by an ecosystem services model by showing data through immersive VR, creating a sense of presence in data, and augmenting the ecosystem services model with other geospatial datasets required to interpret data. This project focuses on immersive VR for analysing geospatial data about ecosystem services following a User Centered Design process [6]. Since different tasks impact the effectiveness of visualizations, experts from different domains performing different tasks with a dataset have their own visualization requirements.

Immersive analytics systems use hardware interfaces to enhance the immersive nature of the experience when analysing data [8]. Immersive visualization systems for analysing tradeoffs among ecosystem services could not be found in the reviewed literature, however, geospatial data can be visualized with immersive systems, such as OpenGEOSys [7]. Model simulation data can be visualized to evaluate whether results are sensible, however, OpenGEOSys was designed for expert analysts. OpenGEOSys was used with scientists from other disciplines on a display wall. However, little work investigated how visualization fits into the workflow of different stakeholders and which visualizations were suitable for users with different backgrounds to communicate data.

This research develops *Immersive ESS Visualizer*, a VR prototype for ecosystem services tradeoff analysis. The prototype needs to satisfy land management scientists' requirements while being understandable to users without the same analytical skills. The prototype's effectiveness will be investigated through these research questions:

RQ1 How can immersive visualizations improve the *comparison* of the impact of land use change for multiple ecosystem services?

RQ2 How effective is immersive visualization for facilitating *communication* between different stakeholder groups analysing the impact of land use change on ecosystem services?

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2 IMMERSIVE ANALYTICS FOR ECOSYSTEM SERVICES

The research questions are being investigated through interviews and focus groups with user groups to inform the development of a novel VR system for visualizing ecosystem service tradeoffs. Interviews and focus groups were approved by the Victoria University of Wellington Human Ethics Committee. 8 participants were recruited for interviews three of whom identified as LUCI developers, three as analysts, and two as university students. Two of the analysts identified as business users. Information was collected about experts' experiences in using LUCI, how their results are communicated, and any problems encountered. After transcription, data were analyzed to find common themes.

Problems encountered with existing solutions: When analysing ecosystem services with LUCI, participants encountered some issues which made their analysis more challenging. For example, one expert stated: *"It took me a long time to figure out how to incorporate temporal change in that output. Because it didn't give me an actual amount of crops that could be created, there was no number."* Another expert participant indicated that the presentation order could cause communication challenges when presenting results to farmers.

Suggested Features: Participants suggested the following features: Temporal data visualization, numerical visualization for rivers and ecosystem services, visualization of land use changes, interaction through flying and walking, directing the users' attention to salient data, different colour schemes, automatically generating graphs, and interactive presentation order.

An initial working software prototype was produced showing ecosystem services tradeoff maps draped over a Digital Elevation Model (DEM). A focus group with four expert participants gathered further information about interview topics, as well as responses to paper prototypes and the working prototype. Follow up interviews elicited information about how visualization designs could assist an analysis with LUCI, and provided feedback to improve the design.

In Fig. 1a the tradeoff layer shows the N.PRESERVE attribute. The purple area has one ecosystem service to preserve, and the grey area has two. The layer chooser on the left hand controller shows tradeoff maps as images labelled with the services. The user selects images, with the right hand laser pointer, and pulls the trigger to change layer Fig. 1b. The attribute list includes ecosystem services, and categorizations from LUCI. Attributes are selected by pointing a controller and pressing the trigger Fig. 1c. The legend is held by the player showing the selected attribute and colour scale for the layer. Users navigate by flying around the map to inspect locations of interest. The left hand controller operates a filter, attribute values smaller than a threshold are transparent Fig. 1a. The working prototype received several positive comments. *"It's the way that you can manipulate and see your different information, which is going to be really great in VR, in 3D, yeah"* One participant's response to viewing layers in VR. *"Yeah, I think I would use it. I think this would be useful for data checking."* Developing filtering methods for showing the underlying data over smaller areas and reducing the complexity of showing values from tradeoff maps were concerns. When presented with a concept for filtering data. *"Yeah, I like that. I think that's actually one of the more useful things."* Another participant suggested inspecting a smaller subset of data. Planned key features of the tool are the comparison of tradeoff maps, visualization of input data and filtering to find the most important information. This will be achieved through features to compare layers, show river information, show input data, change colour scales, filter individual values inside tradeoff maps and add map locations. A concept for side by side comparison received a positive response. *"I think I think this way is easier to contrast. Yeah, that's interesting. Side by side, yeah, there's a very good idea"*

An evaluation of the design will measure the effectiveness of visualizations for completing tasks identified from the interviews and focus group. Case studies will provide more qualitative information

about the effectiveness of the VR visualization system. The case study will be focused on the Mangatarere catchment area in NZ, and will allow for more in-depth analysis than a laboratory evaluation.

The software architecture was developed by analysing the user requirements and related literature for visualizing geospatial data (Fig. 2). VTK was chosen as a visualization library due to the use of VTK in other geospatial visualization projects [1, 2] and the support for geographical file formats and layers. VTK was interfaced with the Unity game engine through Unity's native plugin interface. Tradeoff maps are processed from GeoTIFF rasters using VTK to align with the DEM based on the extents, bounds, and data spacing of the image. Unity generates textures from data. Shaders were written in HLSL to recolour or filter based on the raster attribute tables. Filtering to reduce the complexity of the tradeoff information is partially implemented.

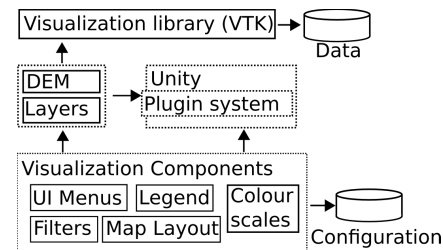


Figure 2: Architecture diagram describing system components

Immersive ESS Visualizer will allow users to compare tradeoffs among ecosystem services and allow both experts and non-experts to explore land use changes in VR and communicate through visualization. Novel features of *Immersive ESS Visualizer* include the ability to visualize tradeoffs and filter data based on the results of the ecosystem services model. The following questions relate to the evaluation and the design of the case study: *How would a case study be formulated to be of interest to the VR community? How would effectiveness for VR be best integrated into a case study?*

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